WORKSHEET

Chapter 5: Position, velocity and acceleration

1 The velocities of a number of particles are listed below. Find the magnitude of the velocity in each case.

a $\mathbf{v} = 4\mathbf{i} + 3\mathbf{j}$

b $\mathbf{v} = -6\mathbf{i} - 5\mathbf{j}$

c $\mathbf{v} = 2\mathbf{i} + 7t\mathbf{j}$

d $\mathbf{v} = 3t\mathbf{i} - t\mathbf{j}$

e $\mathbf{v} = 2\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j}$

f $\mathbf{v} = \frac{4}{t}\mathbf{i} + 3\sqrt{2t}\,\mathbf{j}$

2 The position ($\mathbf{r}$) of a number of points at time t is given below. Find the velocity for each point.

a $\mathbf{r} = 5\mathbf{i} - 3t\mathbf{j}$

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b $\mathbf{r} = \cos(4t)\mathbf{i} + \sin(4t)\mathbf{j}$

c $\mathbf{r} = 3e^{t}\mathbf{i} + 4e^{-2t}\mathbf{j}$

d $\mathbf{r} = 5\sin(2t)\mathbf{i} - 5\cos(2t)\mathbf{j}$

e $\mathbf{r} = 4t\mathbf{i} + 2t^{2}\mathbf{j} - 9t\mathbf{k}$

f $\mathbf{r} = \sin(t)\mathbf{i} - \cos(t)\mathbf{j} + 4t\mathbf{k}$

3 Calculate the magnitude of each velocity found in Question **2**.

a

b

c

d

e

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f

4 Find the acceleration for each point described in Question **2**.

a

b

c

d

e

f

5 Calculate the magnitude of each acceleration found in Question **4**.

a

b

c

d

e

f

6 Given the following information regarding the velocity and position of a particle at time t, determine an expression for the position of the particle in each case.

a $\mathbf{v} = 6t\mathbf{i} - 4\mathbf{j}$ and $\mathbf{r} = 3\mathbf{i} + 3\mathbf{j}$ at $t = 1$

b $\mathbf{v} = 3\cos(t)\mathbf{i} - (4\sin(3t) - 3)\mathbf{j}$ and $\mathbf{r} = 2\mathbf{i} + \mathbf{j}$ at $t = 0$

c $\mathbf{v} = \dfrac{3}{2\sqrt{t}}\mathbf{i} + 10t\mathbf{j}$ and $\mathbf{r} = 3\mathbf{i} + 4\mathbf{j}$ at $t = 1$

d $\mathbf{v} = 12e^{4t}\mathbf{i} + 4t\mathbf{j}$ and $\mathbf{r} = 4\mathbf{i} - 5\mathbf{j}$ at $t = 0$

e $\mathbf{v} = (10t + 7)\mathbf{i} + (12t^2 - 6)\mathbf{j}$ and $\mathbf{r} = 10\mathbf{i} + \mathbf{j}$ at $t = 1$

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7 For each situation in Question **6**, determine an expression for the acceleration of the particle in each case.

a

b

c

d

e

8 A particle has an acceleration $\mathbf{a} = 2\mathbf{i} + 6\mathbf{j}$. The particle has an initial velocity $\mathbf{v}(0) = \mathbf{i} - \mathbf{j}$ and initial position $\mathbf{r}(0) = -2\mathbf{i} + 3\mathbf{j}$. Find an expressions for its velocity and its position at any time, t.

9 A particle travels with a velocity $\mathbf{v} = t^2\mathbf{i} + (e^t + 1)\mathbf{j}$. The particle has an initial position $\mathbf{r}(0) = \mathbf{i} + 3\mathbf{j}$. Determine its position after 1 second.

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Answers

1 **a** 5

b $\sqrt{61}$

c $\sqrt{49t^2 + 4}$

d $\sqrt{10}|t|$

e 2

f $\frac{1}{t}\sqrt{16 + 18t^3}$

2 **a** $\mathbf{v} = -3\mathbf{j}$

b $\mathbf{v} = -4\sin(4t)\mathbf{i} + 4\cos(4t)\mathbf{j}$

c $\mathbf{v} = 3e^t\mathbf{i} - 8e^{-2t}\mathbf{j}$

d $\mathbf{v} = 10\cos(2t)\mathbf{i} + 10\sin(2t)\mathbf{j}$

e $\mathbf{v} = 4\mathbf{i} + 4t\mathbf{j} - 9\mathbf{k}$

f $\mathbf{v} = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + 4\mathbf{k}$

3 **a** 3

b 4

c $e^{-2t}\sqrt{9e^{6-t} + 64}$

d 10

e $\sqrt{16t^2 + 97}$

f $\sqrt{17}$

4 **a** $\mathbf{a} = \mathbf{0}$

b $\mathbf{a} = -16\cos(4t)\mathbf{i} - 16\sin(4t)\mathbf{j}$

c $\mathbf{a} = 3e^t\mathbf{i} + 16e^{-2t}\mathbf{j}$

d $\mathbf{a} = -20\sin(2t)\mathbf{i} + 20\cos(2t)\mathbf{j}$

e $\mathbf{a} = 4\mathbf{j}$

f $\mathbf{a} = -\sin(t)\mathbf{i} + \cos(t)\mathbf{j}$

5 **a** 0

b 16

c $e^{-2t}\sqrt{9e^{6-t} + 256}$

d 20

e 4

f 1

6 **a** $\mathbf{r} = 3t^2\mathbf{i} - (4t - 7)\mathbf{j}$

b $\mathbf{r} = [3\sin(t) + 2]\mathbf{i} + [4\cos(t) - 3t]\mathbf{j}$

c $\mathbf{r} = 3\sqrt{t}\,\mathbf{i} + (5t^2 - 1)\mathbf{j}$

d $\mathbf{v} = (3e^{4t} + 1)\mathbf{i} + (2t^2 - 5)\mathbf{j}$

e $\mathbf{v} = (5t^2 + 7t - 2)\mathbf{i} + (4t^3 - 6t + 3)\mathbf{j}$

7 **a** $\mathbf{v} = 6\mathbf{i}$

b $\mathbf{v} = -3\sin(t)\mathbf{i} - 4\cos(t)\mathbf{j}$

c $\mathbf{v} = \frac{-3}{4\sqrt{t^3}}\mathbf{i} + 10\mathbf{j}$

d $\mathbf{v} = 48e^{4t}\,\mathbf{i} + 4\mathbf{j}$

e $\mathbf{v} = 10\mathbf{i} + 24t\mathbf{j}$

8 $\mathbf{v}(t) = (2t + 1)\mathbf{i} + (3t^2 - 1)\mathbf{j}$;
$\mathbf{r}(t) = (t^2 + t - 2)\mathbf{i} + (t^3 - t + 3)\mathbf{j}$

9 $\mathbf{r}(t) = \left(\frac{1}{3}t^3 + 1\right)\mathbf{i} + (e^t + t + 2)\mathbf{j}$;
$\mathbf{r}(1) = \frac{4}{3}\mathbf{i} + (e + 3)\mathbf{j}$

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